Security group rules for Azure

Cloud Manager 3.6

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Security group rules for Azure

Cloud Manager creates Azure security groups that include the inbound and outbound rules that Cloud Manager and Cloud Volumes ONTAP need to operate successfully. You might want to refer to the ports for testing purposes or if you prefer your to use own security groups.

Rules for Cloud Manager

The security group for Cloud Manager requires both inbound and outbound rules.

Inbound rules for Cloud Manager

The source for inbound rules in the predefined security group is 0.0.0.0/0.

Protocol	Port	Purpose
SSH	22	Provides SSH access to the Cloud Manager host
HTTP	80	Provides HTTP access from client web browsers to the Cloud Manager web console
HTTPS	443	Provides HTTPS access from client web browsers to the Cloud Manager web console

Outbound rules for Cloud Manager

The predefined security group for Cloud Manager opens all outbound traffic. If that is acceptable, follow the basic outbound rules. If you need more rigid rules, use the advanced outbound rules.

Basic outbound rules

The predefined security group for Cloud Manager includes the following outbound rules.

Protocol	Port	Purpose
All TCP	All	All outbound traffic
All UDP	All	All outbound traffic

Advanced outbound rules

If you need rigid rules for outbound traffic, you can use the following information to open only those ports that are required for outbound communication by Cloud Manager.

The source IP address is the Cloud Manager host.

Service	Protocol	Port	Destination	Purpose
Active Directory	TCP	88	Active Directory forest	Kerberos V authentication
	TCP	139	Active Directory forest	NetBIOS service session
	TCP	389	Active Directory forest	LDAP
	TCP	445	Active Directory forest	Microsoft SMB/CIFS over TCP with NetBIOS framing
	TCP	464	Active Directory forest	Kerberos V change & set password (SET_CHANGE)
	TCP	749	Active Directory forest	Active Directory Kerberos V change & set password (RPCSEC_GSS)
	UDP	137	Active Directory forest	NetBIOS name service
	UDP	138	Active Directory forest	NetBIOS datagram service
	UDP	464	Active Directory forest	Kerberos key administration
API calls and AutoSupport	HTTPS	443	Outbound internet and ONTAP cluster management LIF	API calls to AWS and ONTAP, and sending AutoSupport messages to NetApp
API calls	TCP	3000	ONTAP cluster management LIF	API calls to ONTAP
DNS	UDP	53	DNS	Used for DNS resolve by Cloud Manager

Rules for Cloud Volumes ONTAP

The security group for Cloud Volumes ONTAP requires both inbound and outbound rules.

Inbound rules for single node systems

Priority	Name	Port	Protocol	Source	Destination	Action	Description
1000	inbound_ssh	22	TCP	Any	Any	Allow	SSH access to the IP address of the cluster management LIF or a node management LIF
1001	inbound_http	80	TCP	Any	Any	Allow	HTTP access to the System Manager web console using the IP address of the cluster management LIF
1002	inbound_111_tcp	111	TCP	Any	Any	Allow	Remote procedure call for NFS
1003	inbound_111_udp	111	UDP	Any	Any	Allow	Remote procedure call for NFS
1004	inbound_139	139	TCP	Any	Any	Allow	NetBIOS service session for CIFS

Priority	Name	Port	Protocol	Source	Destination	Action	Description
1005	inbound_161-162 _tcp	161-162	TCP	Any	Any	Allow	Simple network management protocol
1006	inbound_161-162 _udp	161-162	UDP	Any	Any	Allow	Simple network management protocol
1007	inbound_443	443	TCP	Any	Any	Allow	HTTPS access to the System Manager web console using the IP address of the cluster management LIF
1008	inbound_445	445	TCP	Any	Any	Allow	Microsoft SMB/CIFS over TCP with NetBIOS framing
1009	inbound_635 _tcp	635	TCP	Any	Any	Allow	NFS mount
1010	inbound_635 _udp	635	TCP	Any	Any	Allow	NFS mount
1011	inbound_749	749	TCP	Any	Any	Allow	Kerberos
1012	inbound_2049_tcp	2049	TCP	Any	Any	Allow	NFS server daemon
1013	inbound_2049_udp	2049	UDP	Any	Any	Allow	NFS server daemon
1014	inbound_3260	3260	TCP	Any	Any	Allow	iSCSI access through the iSCSI data LIF
1015	inbound_4045-4046_tcp	4045-4046	TCP	Any	Any	Allow	NFS lock daemon and network status monitor
1016	inbound_4045-4046_udp	4045-4046	UDP	Any	Any	Allow	NFS lock daemon and network status monitor
1017	inbound_10000	10000	TCP	Any	Any	Allow	Backup using NDMP
1018	inbound_11104-11105	11104-11105	TCP	Any	Any	Allow	SnapMirror data transfer
3000	inbound_deny _all_tcp	Any	TCP	Any	Any	Deny	Block all other TCP inbound traffic
3001	inbound_deny _all_udp	Any	UDP	Any	Any	Deny	Block all other UDP inbound traffic
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow	Inbound traffic from within the VNet
65001	AllowAzureLoad BalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow	Data traffic from the Azure Standard Load Balancer
65500	DenyAllInBound	Any	Any	Any	Any	Deny	Block all other inbound traffic

Inbound rules for HA systems

HA systems have less inbound rules than single node systems because inbound data traffic goes through the Azure Standard Load Balancer. Because of this, traffic from the Load Balancer should be open, as shown in the "AllowAzureLoadBalancerInBound" rule.

Priority	Name	Port	Protocol	Source	Destination	Action	Description
100	inbound_443	443	Any	Any	Any	Allow	HTTPS access to the System Manager web console using the IP address of the cluster management LIF
101	inbound_111_tcp	111	Any	Any	Any	Allow	Remote procedure call for NFS
102	inbound_2049_tcp	2049	Any	Any	Any	Allow	NFS server daemon
111	inbound_ssh	22	Any	Any	Any	Allow	SSH access to the IP address of the cluster management LIF or a node management LIF
121	inbound_53	53	Any	Any	Any	Allow	DNS and CIFS
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow	Inbound traffic from within the VNet
65001	AllowAzureLoad BalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow	Data traffic from the Azure Standard Load Balancer
65500	DenyAllInBound	Any	Any	Any	Any	Deny	Block all other inbound traffic

Outbound rules for Cloud Volumes ONTAP

The predefined security group for Cloud Volumes ONTAP opens all outbound traffic. If that is acceptable, follow the basic outbound rules. If you need more rigid rules, use the advanced outbound rules.

Basic outbound rules

The predefined security group for Cloud Volumes ONTAP includes the following outbound rules.

Protocol	Port	Purpose
All TCP	All	All outbound traffic
All UDP	All	All outbound traffic

Advanced outbound rules

If you need rigid rules for outbound traffic, you can use the following information to open only those ports that are required for outbound communication by Cloud Volumes ONTAP.

The source is the interface (IP address) on the Cloud Volumes ONTAP system.

Service	Protocol	Port	Source	Destination	Purpose
Active Directory	TCP	88	Node management LIF	Active Directory forest	Kerberos V authentication
	UDP	137	Node management LIF	Active Directory forest	NetBIOS name service
	UDP	138	Node management LIF	Active Directory forest	NetBIOS datagram service
	TCP	139	Node management LIF	Active Directory forest	NetBIOS service session
	TCP	389	Node management LIF	Active Directory forest	LDAP
	TCP	445	Node management LIF	Active Directory forest	Microsoft SMB/CIFS over TCP with NetBIOS framing
	TCP	464	Node management LIF	Active Directory forest	Kerberos V change & set password (SET_CHANGE)
	UDP	464	Node management LIF	Active Directory forest	Kerberos key administration
	TCP	749	Node management LIF	Active Directory forest	Kerberos V change & set Password (RPCSEC_GSS)
	TCP	88	Data LIF (NFS, CIFS)	Active Directory forest	Kerberos V authentication
	UDP	137	Data LIF (NFS, CIFS)	Active Directory forest	NetBIOS name service
	UDP	138	Data LIF (NFS, CIFS)	Active Directory forest	NetBIOS datagram service
	TCP	139	Data LIF (NFS, CIFS)	Active Directory forest	NetBIOS service session
	TCP	389	Data LIF (NFS, CIFS)	Active Directory forest	LDAP
	TCP	445	Data LIF (NFS, CIFS)	Active Directory forest	Microsoft SMB/CIFS over TCP with NetBIOS framing
	TCP	464	Data LIF (NFS, CIFS)	Active Directory forest	Kerberos V change & set password (SET_CHANGE)
	UDP	464	Data LIF (NFS, CIFS)	Active Directory forest	Kerberos key administration
	TCP	749	Data LIF (NFS, CIFS)	Active Directory forest	Kerberos V change & set password (RPCSEC_GSS)
DHCP	UDP	68	Node management LIF	DHCP	DHCP client for first-time setup

Service	Protocol	Port	Source	Destination	Purpose
DHCPS	UDP	67	Node management LIF	DHCP	DHCP server
DNS	UDP	53	Node management LIF and data LIF (NFS, CIFS)	DNS	DNS
NDMP	TCP	18600–18699	Node management LIF	Destination servers	NDMP copy
SMTP	TCP	25	Node management LIF	Mail server	SMTP alerts, can be used for AutoSupport
SNMP	TCP	161	Node management LIF	Monitor server	Monitoring by SNMP traps
	UDP	161	Node management LIF	Monitor server	Monitoring by SNMP traps
	TCP	162	Node management LIF	Monitor server	Monitoring by SNMP traps
	UDP	162	Node management LIF	Monitor server	Monitoring by SNMP traps
SnapMirror	TCP	11104	Intercluster LIF	ONTAP intercluster LIFs	Management of intercluster communication sessions for SnapMirror
	TCP	11105	Intercluster LIF	ONTAP intercluster LIFs	SnapMirror data transfer
Syslog	UDP	514	Node management LIF	Syslog server	Syslog forward messages

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